AARAV DESHMUKH
Robotics Mechanical Design Engineer | SOLIDWORKS | ROS 2 | Kinematics | GD&T | AMR
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https://linkedin.com/in/aarav-demo | https://github.com/aarav-demo/engineering-portfolio

SUMMARY

Mechanical engineering graduate with dated internship and competition hardware experience. Mechanisms, mechatronics & autonomous systems.

EDUCATION

Pune Institute of Technology | B.Tech in Mechanical Engineering | Jun 2026
CGPA: 8.6 / 10.0

INTERNSHIPS

Motion Works - Robotics Design Intern | Pune, Maharashtra | Jan 2026 - Jul 2026
- Designed an Autonomous Mobile Robot (AMR) chassis in SOLIDWORKS with Geometric Dimensioning and Tolerancing (GD&T), reducing chassis mass by 18%.
- Validated planetary gearbox design and actuator sizing in MATLAB and Simulink using inverse kinematics, reducing actuator torque by 12 Nm.
- Integrated Robot Operating System (ROS 2) and Unified Robot Description Format (URDF) for a chassis model in Gazebo using Python and C++, reducing simulation cycle time by 25%.

ENGINEERING PROJECTS

Robocon Team - Mechanism Lead | Pune, Maharashtra | Aug 2025 - Dec 2025
- Optimized End-of-Arm Tooling (EOAT) in SOLIDWORKS using Finite Element Analysis (FEA), reducing gripper mass by 220 g.
- Fabricated a sheet metal LiDAR sensor bracket in SOLIDWORKS, achieving a positional tolerance of 0.05 mm.

TECHNICAL SKILLS

SOLIDWORKS | Robot Operating System (ROS 2) | Kinematics | Geometric Dimensioning and Tolerancing (GD&T) | Autonomous Mobile Robot (AMR)

CERTIFICATIONS

Engineering design and hardware safety workshop | Feb 2026